

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Part-III Examination, 2014-15

Paper No. Paper-X: Practical and Programming in C (Old Course)

Time: 3 Hours

Maximum Marks: 70

*Instructions: Question No. 1 is compulsory. Answer **five** questions in all, selecting one question from each Unit. Symbols have their usual meanings.*

Question 1 (Compulsory). Answer all parts briefly: [20]

(a) Write the name of a function provided for output on the screen in C. Write a C statement to display the name of your university on the screen. [2]

(b) Evaluate the following expressions in C, assuming `int a = 1, b = 2, c = 3, d = 4;`:

(i) `b * c / d`

(ii) `a + c % d`

[2]

(c) Write any two logical operators used in C and explain their meaning with examples. [2]

(d) Which of the following are **not** reserved words (keywords) in C?

`break, continue, goto, exit, int`

[2]

(e) Define a variable. Choose the **valid** variable names from the following list:

`A, -BSC, paper-no-x, _student_id, name26, 101name`

[2]

(f) Write equivalent logical expressions in C **without** using negation (!):

(i) `!(a > b)`

(ii) `!(a == b)`

[2]

(g) What will be the output of the following C program?

```
#include <stdio.h>
int main() {
    int i = 5;
    int *j = &i;
    printf("%d\n", *j);
    return 0;
}
```

[2]

- (h) What will be the output of the following C program?

```
#include <stdio.h>
int main() {
    int a[] = {0, 1, 2, 3, 4};
    int i = 0, *p;
    p = &a[0];
    while (i < 5) {
        printf("%d\n", *p);
        p++;
        i++;
    }
    return 0;
}
```

[2]

- (i) What will be the output of the following C program?

```
#include <stdio.h>
struct book {
    char name[30];
    char author[30];
    int year;
};
int main() {
    struct book b1 = {"statistics", "A.B.C", 2002};
    printf("%s\n", b1.name);
    printf("%s\n", b1.author);
    printf("%d\n", b1.year);
    return 0;
}
```

[2]

- (j) Write a small C program that uses an *array of structures* to store data for 3 students (name and marks) and print them. [2]

UNIT I

Question 2.

- (a) Describe the basic data types in C along with their memory sizes and value ranges. Give one example of each data type. [6.5]
- (b) What is a constant in C? Describe different types of constants with examples. [7]

Question 3.

- (a) What are the functions of the increment (++) and decrement (--) operators in C? Explain each with an example. [6]
- (b) What will be the output of the following program? Explain your logic.

```
#include <stdio.h>
int main() {
    int a, b = 0, c = 0;
    a = ++b + c++;
    printf("%d %d %d\n", a, b, c);
    return 0;
}
```

[3]

(c) Evaluate the following expressions in C, assuming `int a = 2, b = -3, c = 5, d = -7;`:

(i) `a / b + c`

(ii) `7 + c * -d`

(iii) `a *= b *= c *= 1`

[4.5]

UNIT II

Question 4.

- (a) Describe the `while` loop structure in C with syntax and a suitable example. [6.5]
- (b) Can an `if-else` statement always be expressed as a `switch` statement? Justify your answer with an example. [7]

Question 5.

- (a) Explain arrays in C. You are given a 5×4 matrix of real numbers. Write a C program to read this matrix and calculate and print its column means. [7]
- (b) Define automatic (local) and static variables in C. Write an example to explain their difference in behaviour across function calls. [6.5]

UNIT III

Question 6.

- (a) Describe pointers and their declaration in C. Give an example of accessing and modifying a variable through its pointer. [6.5]
- (b) Define an array and describe the relationship between arrays and pointers in C with a suitable example. [7]

Question 7.

- (a) Define a function and explain its uses in a C program. Write a C program that reads n real numbers from the screen and computes their standard deviation using a separate function. [7]
 - (b) What is recursion? How is it different from iteration? Write a C program to compute the factorial of a given positive integer using recursion. [6.5]
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UNIT IV

Question 8.

- (a) Describe structures in C. Write a C program to store the name, age, and marks of 30 students in a file using structures. [7]
- (b) Explain unions in C. Describe the similarities and differences between structures and unions. [6.5]

Question 9.

- (a) What is a linked list? Write a C program to create a linear linked list of integers by appending elements at the end and then print the list. [7]
- (b) Write a C program to read 50 real numbers from the screen and find their variance using a separate function. [6.5]